Java Homework2

Task 1

Write a Java method to count all words in a string.
Test Data:
Input the string: I love Java Programming Language
Expected Output:

```
Number of words in the string: 5
```

Task2

Write a program that lets the user perform arithmetic operations on two numbers. Your program must be menu driven, allowing the user to select the operation (+, -, *, / or %) and input the numbers. Furthermore, your program must consist of following functions:
1. Function showMenu: This function shows the options to the user and explains how to enter data.
2. Function add: This function accepts two number as parameters and returns sum.
3. Function subtract: This function accepts two number as parameters and returns their difference.
4. Function multiply: This function accepts two number as parameters and returns product.
5. Function divide: This function accepts two number as parameters and returns quotient.
6. Function modulus: This function accepts two number as parameters and returns a remainder.
A user can perform arithmetic operation many times.

Example:

MENU
1: Add
2: Subtract
3: Multiply
4: Divide
5: Modulus
6: Exit
Enter your choice :1
Enter a number1: 12
Enter a number2: 24
The result is 36

Do you want to perform another operation (Y or N)? Y

MENU
1: Add

2: Subtract
3: Multiply
4: Divide
5: Modulus
6: Exit
Enter your choice :2
Enter a number1: 40
Enter a number2: 25
The result is 15

Task3

Write a Java method to check whether a string is a valid password

Password rules:
A password must have at least eight characters.
A password consists of only letters and digits.
A password must contain at least two digits.

Expected Output:

```
1. A password must have at least eight characters.
2. A password consists of only letters and digits.
3. A password must contain at least two digits
Input a password (You are agreeing to the above Terms and
Conditions.): abcd1234
Password is valid: abcd1234
```

Task4

Write a Java method (takes a number n as input) to displays an n-by-n matrix with random numbers

Expected Output:

```
Input a number: 10
1 0 0 1 1 0 0 0 1 1
0 0 1 0 1 0 1 0 0 0
0 1 0 1 0 0 0 0 0 1
1 1 1 0 0 0 0 1 1 1
1 1 0 1 1 1 0 1 0 0
1 0 0 0 1 1 0 0 0 0
0 0 1 0 0 0 0 1 1 1
1 1 0 1 0 1 0 0 1 0
0 0 1 0 0 0 0 1 1 0
```

1 1 1 0 0 1 1 1 1 0

Task5

Write a program to implement the Number Guessing Game. In this game the computer chooses a random number between 1 and 6, and the player tries to guess the number in as few attempts as possible. Each time the player enters a guess, the computer tells him whether the guess is too high, too low, or right. The player can play the game many times as he or she wants.

The output:

```
Guess My Number Game
Enter a guess between 1 and 6 : 5
Too high! The random number is 3
Do you want to play (Y or N) ? Y
Enter a guess between 1 and 6 : 1
Too low! The random number is 2
Do you want to play (Y or N) ? Y
Enter a guess between 1 and 6 : 2
Correct! You got it in 3 guesses!
```